EXP 4: Artificial Neural Network using Backpropagation

CODE

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
data = load_breast_cancer()
X, y = data.data, data.target.reshape(-1, 1)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
np.random.seed(42)
W1 = np.random.randn(X_train.shape[1], 16) * 0.1
b1 = np.zeros((1, 16))
W2 = np.random.randn(16, 1) * 0.1
b2 = np.zeros((1, 1))

def sigmoid(z):
    z = np.clip(z, -500, 500)
    return 1 / (1 + np.exp(-z))
loss_history = []
accuracy_history = []
lr = 0.05
for epoch in range(1000):
```

```
    A1 = sigmoid(X_train @ W1 + b1)
    A2 = sigmoid(A1 @ W2 + b2)
    eps = 1e-8
    loss = -np.mean(
        y_train * np.log(A2 + eps) +
        (1-y_train) * np.log(1-A2 + eps)
    )
    loss_history.append(loss)
    m = len(X_train)
    dZ2 = (A2 - y_train) / m
    dW2 = A1.T @ dZ2
    db2 = np.sum(dZ2, axis=0, keepdims=True)

    dZ1 = (dZ2 @ W2.T) * A1 * (1-A1)
    dW1 = X_train.T @ dZ1
    db1 = np.sum(dZ1, axis=0, keepdims=True)
    W2 -= lr * dW2
    b2 -= lr * db2
    W1 -= lr * dW1
    b1 -= lr * db1
    predictions = (A2 >= 0.5).astype(int)
    accuracy_history.append(np.mean(predictions == y_train) * 100)
A1_test = sigmoid(X_test @ W1 + b1)
A2_test = sigmoid(A1_test @ W2 + b2)
y_pred = (A2_test >= 0.5).astype(int)
test_accuracy = np.mean(y_pred == y_test) * 100
print("Dataset: Breast Cancer")
print("Training samples:", len(X_train))
print("Testing samples:", len(X_test))
```

```
print("Test Accuracy: {:.2f}%".format(test_accuracy))
plt.figure(figsize=(7, 4))
plt.plot(loss_history)
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("ANN Training Loss")
plt.grid(True)
plt.show()
plt.figure(figsize=(7, 4))
plt.plot(accuracy_history)
plt.xlabel("Epoch")
plt.ylabel("Accuracy (%)")
plt.title("ANN Training Accuracy")
plt.grid(True)
plt.show()
```

OUTPUT

Dataset: Breast Cancer

Training samples: 455

Testing samples: 114

Test Accuracy: 97.37%



